2. Light and Seeing/Design of a Lighting System

The last chapter discussed the nature of light. Light was seen as radiant energy transported in photons that are carried by a wave field. In this chapter, the eye is discussed. Also discussed are some objective factors in the seeing process. The chapter ends by introducing some terms used by illumination engineers.

Behavior of Light

In the last chapter, the origin of light was discussed. This chapter discusses briefly the behavior of light after it leaves the source. Three basic characteristics of light will be discussed. The first characteristic is:

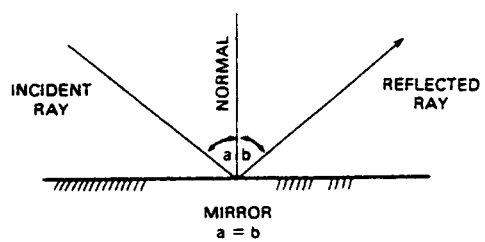
> Light travels in a straight line unless it is modified or redirected by means of a reflecting, refracting, or diffusing medium.

When light travels, it travels in a straight line. When it is incident upon a surface, part of the light is reflected. On a metallic surface, almost 100 percent of the light is reflected; while on a clear glass surface, only a small portion is reflected. The ratio of light reflected from a surface to that incident upon it is called reflectance.

The law of reflectance is simply stated as: The angle of incidence is equal to the angle of reflection. (See Figure 5.2.1.)

Figure 5.2.1

Law of reflection.



Reflections may be of several types; the most common are specular (see Figure 5.2.2), diffuse (see Figure 5.2.3), spread reflection (see Figure 5.2.4), and mixed reflection (see Figure 5.2.5), which is a combination of diffuse and spread reflection.

Figure 5.2.2

Specular reflection.

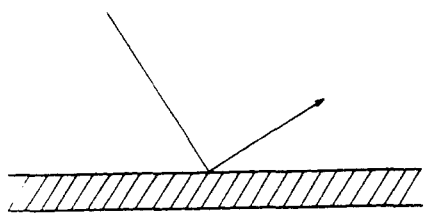


Figure 5.2.3

Diffuse reflection.

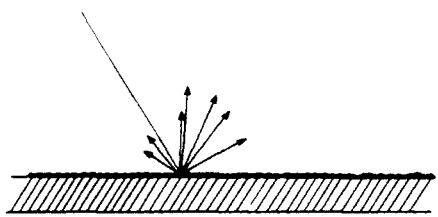


Figure 5.2.4

Spread reflection.

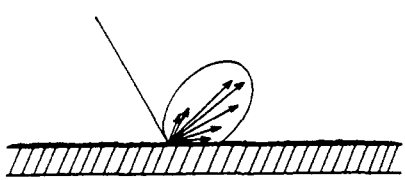
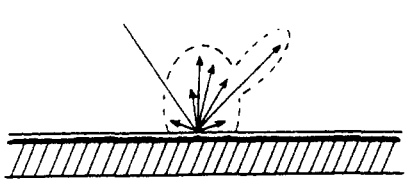


Figure 5.2.5

Mixed reflection.



Refraction is the "bending" of light as it passes from one transparent medium to another. The speed at which light travels through these materials is what makes light bend. The speed of light is consistent in space. However, light has a lesser speed in a transparent medium. In water, light travels 75 percent of its speed compared to a vacuum; in glass, about 67 percent, depending upon the type of glass; in a diamond, about 41 percent. When light emerges from these media, it again travels at its original speed. This concept may be troublesome because from what is known about energy, this may seem like strange behavior. If a bullet is fired through a board, the bullet slows when passing through the board and emerges at a speed less than its incident speed. It loses some of its kinetic energy while interacting with the fibers and splinters in the board. But things are different with light. To understand the behavior of light, the individual photons of light that make up a beam and the interaction between the photons and the molecules they encounter must be considered. Incident photons interact with the electrons of molecules. Orbiting electrons can be thought of as attached to little springs. These electrons will resonate at certain frequencies and can be forced into vibration over the range of frequencies. This range varies for different molecules. In clear glass, for example, the range extends over the entire visible region. When a photon is incident upon a transparent medium such as glass, it is absorbed by a molecule at the surface. An electron in the absorbing molecule is set in vibration at a frequency equal to that of the incident photon. This vibration then causes the emission of the second photon of identical frequency. It is a different but indistinguishable photon. The second photon travels at 186,000 miles per second until it quickly is absorbed by another molecule in the glass, whereupon an electron is set in vibration re-emitting a different but indistinguishable photon of its own. This absorption/re-emission process is not an instantaneous event. Some time is required for the process; and, as a result, the average speed of light through the material is less than 186,000 miles per second. That is, the photon that enters the glass is not the same photon that leaves the glass.

Light bends when it passes obliquely from one medium to another. This is called refraction. It is the slowing of light upon entering the transparent medium that causes the refraction. (See Figure 5.2.6.)

The straight line travel of light can also be altered by a diffusing material. Light traveling through a transparent or translucent material is said to be transmitted, such as light traveling through a clear glass plate. When light leaves the material, it may become diffuse. (See Figure 5.2.7.) The degree of diffusion depends upon the type and density of the material. Most luminaires are made so that the light leaving the luminaire becomes diffuse.

The second characteristic of light is:

> Light waves pass through one another without alteration of either.

That is, a beam of red light will pass directly through a beam of blue light unchanged in direction and color.

Figure 5.2.6

Refraction of light.

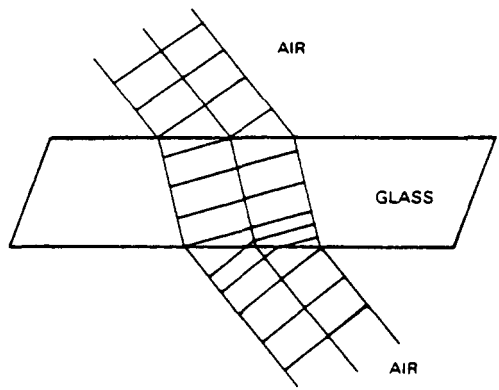
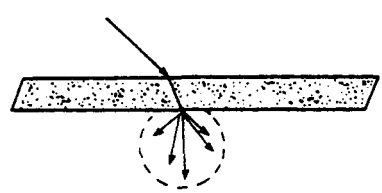


Figure 5.2.7

Diffusing glass.



The third characteristic of light is:

> Light is invisible in passing through space unless some medium, such as dust or water, scatters it in the direction of the eye.

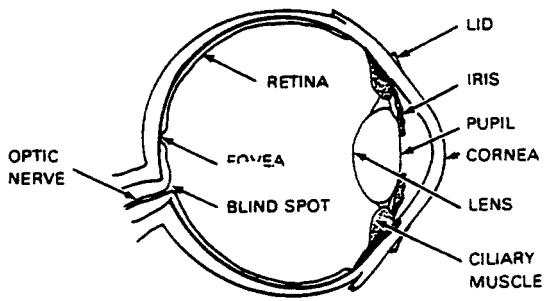
The scattering of light is similar to the phenomenon of resonance in sound and forced vibration. Atoms and molecules behave like tuning forks and selectively scatter waves of the appropriate frequency. A beam of light falls upon an atom and causes the electrons to vibrate. The vibrating electron in turn radiates light in different directions. An example of scattering is a searchlight beam sweeping across the sky at night. Such beams are seen by light being scattered by particles (dust or water droplets) in the atmosphere.

The Human Eye

Figure 5.2.8 shows the structure of the eye. Light passes through the cornea (a protective coating over the front of the eye). Light next passes

Figure 5.2.8

The human eye.



through the pupil, an opening in the iris that can be widened or narrowed to let more or less light in by contractions in the muscles of the iris, the colored portion of the eye. Light passes through the pupil into the lens, a transparent capsule behind the iris whose shape can be changed in order to focus objects at various distances. The lens is controlled by the ciliary muscle, which is ring shaped and changes the curvature of the lens. The light is then focused through the lens into the inner lining of the back of the eyeball, the retina. There the light stimulates receptor cells that transmit the information to the brain via the optic nerve.

More than six million cones and 100 million rods are distributed in the retina. Rods are slim nerve cells--receptors--which are sensitive to low levels of illumination. Rods have no color response. They are found only on the outside of the foveal region, increasing in number with the distance from the fovea. The outer portion of the retina is composed chiefly of rods which do not afford distinct vision but are highly sensitive to movement and flicker. When light strikes a rod, it causes the breakdown of a chemical rhodopsin (visual purple). This photosensitive chemical triggers activity in the optic nerve and, subsequently, in the brain.

Cones are the receptors that make possible the discrimination of fine detail and the perception of color. Cones are insensitive at low levels of illumination. The cones are found mainly near the center of the retina, with the greatest concentration at the fovea. A few cones are mixed with rods all the way to the outer edges of the retina, but the center of the eye is the most color-sensitive portion. The cones also contain a photosensitive chemical that breaks down when struck by light waves.

The eye has the ability to adapt to a wide variety of illumination levels. Adaptation involves a change in the size of the pupils along with photochemical changes in the rods and cones. In dim light--low levels of illumination--the chemicals in the rods and cones are built up faster than they are broken down by light stimulation. The greater the concentration of these chemicals, the lower the visual threshold. Thus, the adaptation to

darkness is a matter of building up a surplus of rhodopsin in the rods and other chemicals in the cones.

The cones adapt quickly in the dark (10 minutes or so), but the rods adapt slowly and continue to adapt even after 30 minutes or more of darkness. These are only rough estimates since the length of time of adaptation depends upon the previous state of adaptation and the magnitude of the change. When completely adapted, the rods are much more sensitive to light than the cones. Thus, to see a dim light in pitch darkness, one should not look directly at it since the center of the eye contains only the less sensitive cones. By looking away from the object, the image will fall on the edge of the retina where the rods are. This manner of viewing affords a higher likelihood of seeing the dim light. Since the rods work in dim light and the cones do not, vision in very dim light is entirely colorless. Although vision is colorless in dim light, the eye becomes relatively sensitive to energy at the blue end of the spectrum and almost blind to red.

Visual acuity is the ability to discriminate the details in the field of vision. The normal field of vision extends approximately 180° in the horizontal plane and 130° in the vertical plane (60° above the horizontal and 70° below). One way the ability to discriminate detail in the field of vision can be measured is by using the familiar eye chart. Standard perfect vision is often called "20/20 vision." If a person stands 20 feet away from the eye chart and sees the material on the chart clearly, he or she is seeing normally and is said to have 20/20 vision. If the person does not see normally, some of the material will be blurred. If a person standing 20 feet away from the chart sees what a person with normal vision sees at 50 feet, the person has 20/50 vision. If a person has 20/10 vision, he sees things 20 feet away as sharply as a person with normal vision sees them at 10 feet.

Part of the retina, the "blind spot," has no visual acuity. This spot is the point at which the nerves of the eye converge to form the optic nerve. The optic nerve extends through the back wall of the eyeball and connects the eye to the brain. People are usually unaware of the blind spot; they compensate for this blind spot in their vision primarily by moving their heads and making use of their other eye.

The four most common causes of defective vision are astigmatism, the inability to bring horizontal lines and vertical lines into focus at the same time; myopia, where objects focus in front of the retina--nearsightedness; hypermetropia, where objects focus behind the retina--farsightedness; and presbyopia, loss of elasticity of the lens with age. All of these visual defects can usually be corrected by properly fitted corrective glasses or lenses.

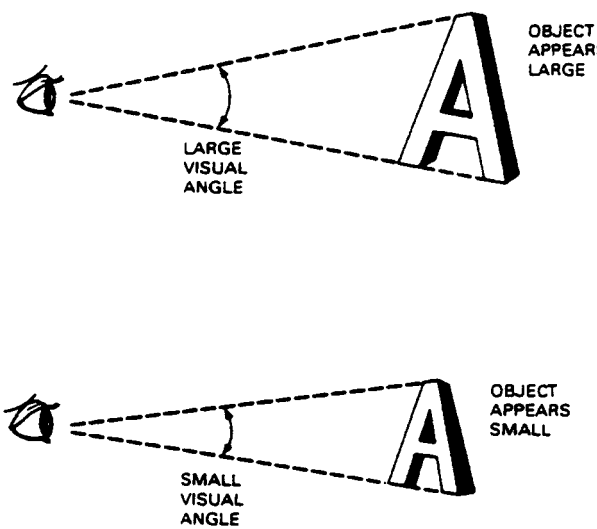
Variables in the Seeing Process

What makes an object easy to see? Investigations have shown that adequate seeing depends upon at least four variables. These are the size of the object, the contrast of the object with its background, the brightness of the object, and the time available to see the light.

The obvious factor in seeing an object is its size. The size of the object depends upon the visual angle. The larger an object in terms of its visual angle--the angle subtended by the object at the eye--the more readily it can be seen. The familiar eye test chart illustrates this principle. The person who brings a small object closer to his eyes in order to see it more clearly is unconsciously making use of the size factor by increasing the visual angle. (See Figure 5.2.9.)

Figure 5.2.9

Size of object--visual angle of object.



Along with the size of the object is visual acuity. Visual acuity, expressed as the reciprocal of the visual angle in minutes, is a measure of the smallest detail that can be seen. Since visual acuity increases markedly with increase in illumination, light is sometimes said to act as a magnifier, making visible small details that could not be seen with less light.

The second factor involved in seeing objects is contrast. Contrast primarily refers to two factors--color contrast and brightness contrast. Color contrast refers to the contrast in color between the object to be seen and its immediate background. For a given set of conditions, visibility is at its highest when the contrast is at a maximum. Black print on white paper is much more visible than the same print on grey paper. (See Figure 5.2.10.) Brightness contrast is the contrast in brightness between the object and its immediate background.

The third factor in seeing is brightness. The luminance or brightness of an object depends upon the intensity of the light striking it and the proportion of that light reflected in the direction of the eye. A white

surface will have a much higher luminance than a black surface receiving the same illumination. However, by adding enough light to a dark surface, it is possible to make it as "bright as the white one." The darker an object or visual task, the greater the illumination necessary for greater illuminance and, under like circumstances, for equal visibility. In addition, the brightness between the object and its immediate background should be approximately the same. Different ratios in brightness between the object and the background can cause problems for the viewer.

The fourth factor in seeing is time. Seeing is not an instantaneous process--it requires time. The eye can see very small detail under very low levels of illumination if sufficient time is allotted and eyestrain is ignored. However, more lighting is required for quick seeing. The time factor is particularly important when the visual object is in motion. High lighting levels actually make moving objects appear to move more slowly and greatly increase their visibility.

Size, luminance, contrast, and time are mutually interrelated and interdependent. Within limits, deficiency in one can be made up by an adjustment in one or more of the others. In most cases, size is a fixed factor of the visual task, with luminance, contrast, and time subject to some degree of modification. Of these, luminance and contrast are usually most directly under the control of the illuminating engineer. Properly employed, they can be of tremendous aid in overcoming unfavorable conditions, small size, and limited time for seeing.

To see the interrelation among the four factors, consider the following:

A. Small objects must have a high contrast to be seen.
B. Low contrast objects must be large in size.
C. As brightness increases, contrast and size can be decreased.
D. When more time is available for seeing objects, the size can be smaller and the contrast lower.
E. In most situations, the object is fixed, contrast is usually fixed, and the time for seeing is fixed; thus, brightness is most often the variable under the control of the engineer.

Terminology Used in the Science of Light

There are some terms related to the science of light with which everyone working with light should be familiar.

Luminous Flux (F). Luminous flux is the total radiant power emitted from a light source that is capable of affecting the sense of sight. Actually, it is more precisely defined as the time rate of flow of light (or luminous energy). In ordinary practice, the time element can be neglected, and luminous flux is commonly considered a definite quality. Luminous fluxis measured in units of a lumen where one lumen is defined as the luminous flux emitted from a 1/60 cm^2 opening in a standard source and included within a solid angle of 1 steradian. The standard source consists of a hollow enclosure maintained at the temperature of solidification of platinum, about 1773°C. A solid angle in steradians is given by:

$$\Omega = \frac{A \cos \theta}{R^2}$$

Where Ω denotes a solid angle, A denotes surface area, and R denotes distance (or radius of a sphere), where R is perpendicular to the surface, A, and θ represents the angle from the center of the sphere to the surface area (A).

A solid angle can be graphically illustrated as in Figure 5.2.11.

Figure 5.2.10

Contrast.

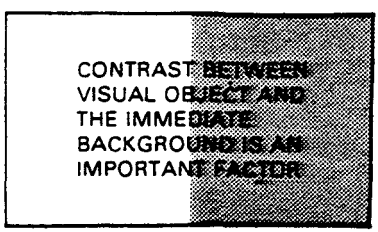
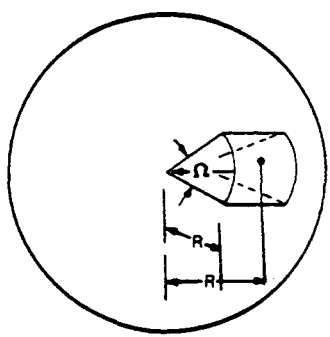


Figure 5.2.11

Definition of a solid angle in steradians.



If $\theta = 0°$, that is, the surface area is perpendicular to the center of the sphere, then $\cos \theta = 1$ and $\Omega = A/R^2$.

In the definition of a lumen, then, one steradian can be defined as the solid angle subtended at the center of a sphere by an area A on its surface that is equal to the square of its radius R. It can be shown that there are 4π steradians in a complete sphere.

$$\Omega = \frac{A}{R^2}$$

$$= \frac{4\pi R^2}{R^2}$$

$$= 4\pi \text{ steradians}$$

A lumen is often defined as the flux falling on a surface one square foot in area, every part of which is one foot from a point source having a luminous intensity of one candela (candlepower) in all directions. Luminous intensity is a term not yet defined.

Luminous Intensity. Luminous intensity of a light source is the luminous flux emitted per solid angle:

$$\text{Intensity} = \text{Flux}/\Omega$$

The unit for intensity is the lumen per steradian called a candela or candlepower, as it was called when the international standard was defined in terms of the quantity of light emitted by the flame of a certain candle. (This initial standard was replaced by the platinum standard.) This standard candle had a luminous intensity in a horizontal direction of approximately one candela. If a light from a candle with the candlepower of 1 fell on a surface of one square foot, where every part of the surface was one foot from the candle, the amount of flux would be one lumen. Note from the above equation that

$$\text{Flux} = \text{Intensity} \times \Omega$$

If the light source is an isotropic source (one which emits light uniformly in all directions), then the total flux emitted would be

$$\text{Flux} = 4\pi \text{ Intensity}$$

since the total solid angle for an isotropic source is 4π steradians.

Example

If a spotlight one foot away from a wall had a bulb with a candlepower of 1 candela and the beam covered an area of 1 square foot of the wall, what would the luminous intensity of the spotlight be?

Solution

Total flux emitted by the 1 candela bulb is

$$\begin{aligned}\text{Flux} &= 4\pi \text{ intensity} \\ &= (4\pi)(1 \text{ candela}) \\ &= 12.56 \text{ lumens} \quad (\text{flux is measured in lumens})\end{aligned}$$

The light is concentrated into a solid angle given by

$$\begin{aligned}\Omega &= \frac{A}{R^2} \\ &= \frac{1 \text{ ft}^2}{(1 \text{ ft})^2} \\ &= 1 \text{ steradian}\end{aligned}$$

And the intensity of the beam is given by

$$\text{Intensity} = \text{Flux}/\Omega$$

$$= \frac{12.56 \text{ lumens}}{1 \text{ steradian}}$$

$$= 12.56 \text{ candela} \quad \text{(intensity is measured in candela)}$$

Note that the unit of intensity, candela (lumens/steradian) and the unit of flux (lumen) are the same dimensionally because the solid angle in steradians is dimensionless. Also note that flux and intensity will be equal when Ω = 1 steradian (i.e., when $A = R^2$).

Illumination. Illumination is the density of luminous flux on a surface area; i.e.,

$$\text{Illumination} = \text{Flux/Area}$$

When flux is measured in lumens and area is measured in square feet, then illumination is expressed in lumens per square feet. The lumens per square feet is sometimes called the foot-candle. If a lumen is defined as the flux falling on a surface area of one square foot, where every part of the surface is one foot from a point source having a luminous intensity of 1 candela (candlepower), then it is obvious that one lumen uniformly distributed over one square foot of surface provides the illumination of one foot-candle. Visually, a foot-candle is the illumination at a point, X, on a surface which is one foot from and perpendicular to a uniform point source of one candela. (See Figure 5.2.12.) In this special case where the incident light is perpendicular to the surface, it can be shown that

$$\text{Illumination} = \frac{\text{Intensity}}{\text{Distance}^2}$$

The relationship

$$\text{Illumination} = \frac{\text{Lumens (flux)}}{\text{Area}}$$

is important and will be used during the discussion of the lumen method of designing a lighting system.

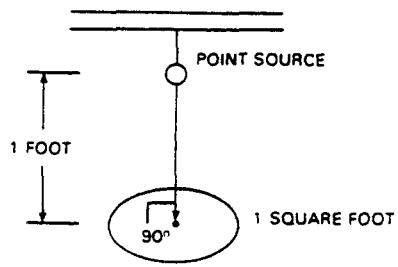
Luminance. Luminance (sometimes called photometric brightness) is a measure of the brightness of a surface, when viewed from a particular direction, emitting or reflecting one lumen per square foot. Luminance is direction-specific and is often measured in foot-lambert units.

Reflectance. Reflectance is a measure of how much light is reflected from a surface. It is the ratio of luminance to illumination.

$$\text{Reflectance} = \frac{\text{Luminance}}{\text{Illumination}}$$

Figure 5.2.12

Foot-candle.



The Measurement of Light

Foot-candle Measurements. Illumination measurements are most commonly made with one of several types of foot-candle meters embodying a light-sensitive, barrier-layer cell. (See Figure 5.2.13.) This type of cell consists essentially of a film of light-sensitive material mounted on a metal-based plate and covered by a very thin translucent layer of metal spattered on its outer surface. Light striking the cell surface causes the semiconducting, light-sensitive material to emit electrons that are picked up by the metal collector in contact with the translucent front electrode. A potential difference is thus set up between the collector and the base plate; and, when a microammeter is connected between them, it measures the current generated by the cell. Since the current is proportional to the intensity of the incident light, the meter can be calibrated to read directly in foot-candles. Portable meters are made in a number of types and with a wide range of sensitivity for various applications. Although portable, light-sensitive cell meters are simple and highly convenient to use, most of them are not designed to be precision instruments. Careful handling and frequent calibration will help to maintain reliability. Ordinary measurements made in the field should not be expected to have an accuracy greater than $\pm$ 5% under the most favorable conditions. In addition, all light-sensitive cells have certain inherent characteristics that the user must understand if he or she is to obtain the best possible results:

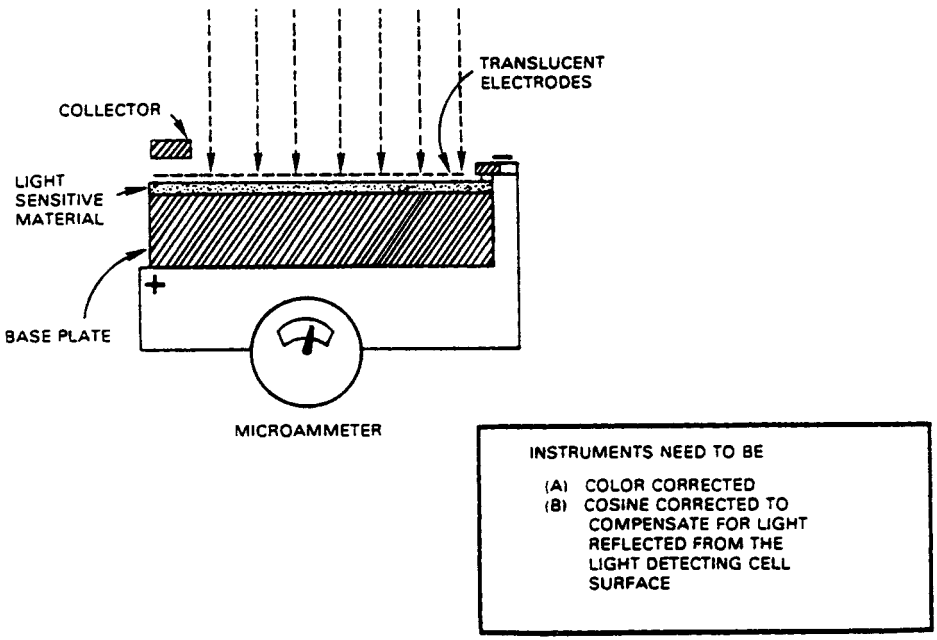
A. The instrument must be color corrected because the response of the light-sensitive cells to the various wavelengths of the visible spectrum is quite different from that of the human eye.

B. They must be cosine corrected; that is, adjusted for the angle of the reflected light.

C. All light-sensitive cells exhibit a certain amount of fatigue; that is, a tendency for the meter indication to drop off slowly over a period of minutes until a constant reading is reached. This effect is most noticeable at high foot-candle values, particularly if the

Figure 5.2.13

Illumination meter.



cell has just been previously in the dark for some time or exposed to a much lower level of illumination. Before any measurements are recorded, therefore, the meter should be given as long an adaptation period as may be necessary at the foot-candle level to be measured. In addition, there is a constant need to have the instrument calibrated.

Brightness or luminance is measured using a photoelectric tube. The instrument is aimed at the surface to be measured, and a lens focuses the image on a small area on the tube which produces a current proportional to luminance. The current is read on a microammeter calibrated in foot-lamberts.

Brightness is also measured by a visual luminance meter. This uses an optical system to bring the eyes of the observer side by side to the surface to be measured and a comparison field inside the meter.

Luminance can be measured with a foot-candle meter providing the reflectance of the surface is known. This is because illuminance equals luminance divided by reflectance.

Reflectance can be measured by a cell-type foot-candle meter. There are two procedures that can be used. The more accurate procedure requires a piece

of matte material at least one foot square, the reflectance of which is known. White blotting paper, at about 80 percent reflectance, is suitable. The blotting paper is placed against the surface to be measured, and the meter is held two to four inches away with the cell facing the paper, reading A. The blotting paper is then removed without moving the meter, and the reading B is noted. The reflectance of the surface is reading B divided by reading A times 0.80. In all measurements of this sort, special care must be taken to maintain all conditions, especially the position of the meter, constant for both readings of a pair.

Light Survey Procedures

"How to Make a Lighting Survey," developed by the Illuminating Engineering Society, provides detailed information on conducting a lighting survey. A set of instructions and a form are included with this information.

The survey includes reporting on the following information:

1. Description of the illuminated area:
    a. Room dimensions
    b. Color
    c. Reflectance
    d. Conditions of room surface
    e. Temperature surrounding the lights
2. Description of the general lighting system:
    a. Quantities
    b. Conditions
    c. Wattages
    d. Lamps
    e. Distribution
    f. Spacings
    g. Mountings
3. Description of any supplementary lighting that might be used.
4. Description of instruments to be used.
5. Illumination measurement:
    a. Operator must be aware not to cast shadows.
    b. Operator must be careful not to reflect additional light from clothing.
    c. Test surfaces should be as close as possible to the working plane. If there is no definite working plane, take measurements on a horizontal plane 30 inches above the floor.
6. Luminance measurements.

Evaluation of Results. Data resulting from a light survey can be used to compare the illumination levels for compliance with the recommended levels; to compare the luminance with compliance luminance levels; to determine luminance ratios for visibility and safety; to determine indications of comfort and pleasantness in the area; to determine deficiencies in the area; and to determine a maintenance schedule--that is, a good housekeeping schedule.